INCH-POUND

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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION
TEST SET, RADIO FREQUENCY POWER
(50 MHz to 40 GHz)

1. SCOPE. This Commercial Item Description (CID) describes the minimum requirements for a Radio Frequency Power Test Set which consists of a dual-channel peak and continuous wave power meter, sensors, and applicable accessories. The equipment, as described in this CID, will have a minimum frequency range of 50 MHz to 40 GHz and will be used for measuring peak power and continuous wave power over selected frequencies.

2. CLASSIFICATION. Equipment covered by this CID shall be commercially available equipment and may be modified to the extent necessary to meet the following description. The equipment shall be Class 3, in accordance with MIL-PRF-28800, except as specified herein.

3. SALIENT CHARACTERISTICS. The equipment shall be capable of operation within the accuracies, limits, and specifications herein.

3.1 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein except as specified herein.

3.1.1 Temperature Operating. The equipment shall conform to its specified performance and accuracy while being operated in a temperature range of 0°C to +50°C.

3.1.2 Temperature Non-operating. The equipment shall conform to its specified performance and accuracy after being stored in an environment in the temperature range of -40°C to +70°C.

3.2 Electrical Power Sources. The equipment shall operate from nominal commercial, military and shipboard power sources of 110/220 volts AC (±10%) at line frequencies of 50 to 60 Hz.

3.3 Reliability. The design of the equipment shall be such that under normal use and operation the equipment does not fail within 2500 hours of operation with a statistical certainty of 95%.

3.4 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. The calibration adjustments, wherever possible, shall be accessible without removal of the equipment case or modules. The calibration interval shall be a period of one year or greater based on an operating time of 2000 hours. No special tools shall be required for alignment or adjustment except those furnished with the equipment.

3.5 Performance Requirements. The equipment shall meet all requirements specified herein, after the warm-up period specified in MIL-PRF-28800, under all combinations of input power conditions, output signal conditions and operating service condition as specified in MIL-PRF-28800.

3.5.1 Meter/Chassis. The peak and continuous wave power meter shall have a minimum of two measurement channels.

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3.5.1.1 Remote Operation. The peak and continuous wave power meter shall have an IEEE-488 interface for remote set up, control, and measurement operation.

3.5.1.2 Frequency Measurement Range. The minimum frequency range of the peak and continuous wave power meter with sensors shall be 50 MHz to 40 GHz.

3.5.1.3 Power Measurement Range. The peak and continuous wave power meter with sensors shall have, as a minimum, a range of -20 dBm to 20 dBm in the pulse and continuous wave modes.

3.5.1.4 Display/Time Base. The peak and continuous wave power meter shall have a digital display for all functions. The display shall provide an analog representation of the pulse being measured. The display shall provide, as a minimum, information on the amplitude and timing of the measured signal.

3.5.1.4.1 Timing Information Display. As a minimum, the following pulse timing information shall be provided by the display: rise time, fall time, pulse width, pulse repetition interval or frequency, pulse duty cycle, and delay.

3.5.1.4.1.1 Time Base. The time base on the display shall be selectable, as a minimum, from 10 ns per division to 20 ms per division. The time base resolution shall be 0.2 ns or better. The time base shall be accurate to less than ± (0.01% of window) ± (1ns).

3.5.1.4.2 Amplitude Information Display. Amplitude information shall be available for display in logarithmic form. The equipment shall be capable of displaying the amplitude measurement range specified herein. As a minimum, the following amplitude information shall be available for display: peak power, average power, pulse top amplitude, pulse bottom amplitude, and overshoot.

3.5.1.4.2.1 Range. The equipment shall include a display range of –99.99 dB to +99.99 dB.

3.5.1.4.2.2 Resolution. The display resolution shall be less than 0.05 dB.

3.5.1.4.2.3 Offset. The equipment shall have an amplitude offset capability to include a range from –99.99 dB to +99.99 dB.

3.5.1.4.3 Triggering.

3.5.1.4.3.1 Trigger Sources. Trigger sources shall include internal, external, and manual. Internal triggering shall be available from any sensor channel. Manual triggering shall be available via the front panel.

3.5.1.4.3.2 Trigger Level. The trigger level range for internal triggering shall include a range of minus 15 dBm to plus 15 dBm. The external trigger range shall be compatible with TTL voltage levels.

3.5.1.4.3.3 Trigger Delay. The equipment shall have a trigger delay capability to include a range from 0.0 to 200 milliseconds.

3.5.1.5 Operating Modes.

3.5.1.5.1 Peak/Continuous Wave (CW) Selection. Selection of the peak or continuous wave modes of measurements, or equivalents, shall be available via the front panel.

3.5.1.5.2 High/Low Video Bandwidth Selection. Selection of high or low video bandwidth, or equivalents, shall be available via the front panel.

3.5.1.6 Power Reference Calibration Source. An internal power reference source, and associated operating mode, shall be provided to allow user-selected automatic calibration of the combination of the sensor(s) and the equipment.

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3.5.1.6.1 Frequency. The power calibration source frequency shall be 1.0 GHz ±5% or 50 MHz ±5%.

3.5.1.6.2 Amplitude. The power reference source amplitude shall be within ± 1.5% at 1 milliwatt (0 dBm). VSWR shall be less than 1.20 at 0 dBm, 50 ohms.

3.5.1.7 Connectors and Cables.

3.5.1.7.1 Power Reference RF Connector. The power calibration source shall have a Type N RF connector mounted on the front panel of the equipment.

3.5.1.7.2 Interconnect cable. The power meter shall come with interconnect cables for connecting sensors to the equipment. Each sensor cable shall be detachable from the sensor head and shall be a minimum of 5 feet (1.5 meters) in length. An interconnect cable shall be provided for each channel.

3.5.1.7.3 Adapters. The equipment shall include adapters to connect sensors, which do not have a compatible Type N RF connector, to the power calibration output.

3.5.1.8.3 Analog Input/External Trigger. The equipment shall have an input connector for external triggering compatible with TTL signals. A connector shall be provided for each measurement channel.

3.5.2 Sensors. Sensors or sensor sets which meet the following characteristics shall be provided for each measurement channel.

3.5.2.1 Sensor Frequency Measurement Range. The minimum frequency range of the peak and continuous wave sensors shall be 50 MHz to 40 GHz.

3.5.2.2 Sensor Power Measurement Range. The sensor(s) shall have, as a minimum, a power range of -20 dBm to 20 dBm in the pulse and continuous wave modes.

3.5.2.2.1 Sensor Calibration Uncertainty. The root-sum-square (RSS) sensor calibration uncertainty shall be less than ±6.5% for the operational frequency range specified herein.

3.5.2.2.2 Sensor VSWR. The sensor VSWR shall be less than 2.0 for the operational frequency range specified herein.

3.5.2.3 Sensor Risetime. Sensor risetime for high video bandwidth setting shall be less than 100 ns. Sensor risetime for low video bandwidth setting shall be less than 10 us.

3.5.2.4 Sensor RF Connection. The sensor RF interconnects shall be Type N, Type APC-3.5mm, or Type K. Equivalents are acceptable.

3.5.2.5 Sensor Calibration Data. Sensors shall be capable of containing calibration data in non-volatile memory which can be electronically loaded directly from the sensor to the meter, without operator intervention after the transfer is initiated.

3.5.3 Self-Test. The equipment shall provide automatic diagnostic self-test information. The information provided shall indicate whether the equipment is operating within the performance specification. If the equipment fails to operate within the performance specification bounds, the information provided by the self-test shall identify the associated lowest replaceable unit.

3.5.4 Physical Characteristics. The test set's component dimensions and weight shall be consistent with current commercial offerings. The equipment shall have the following physical characteristics.

3.5.4.1 Dimensions. The width and depth of the equipment shall be compatible with mounting in an ANSI/EIA-310-D standard equipment rack. The height shall be compatible with five standard height units, 5U. One unit or "U" is defined by ANSI/EIA-310-D to be 1.75 inches or 44.45 mm.

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3.5.4.2 Weight. The total weight of the equipment, excluding accessories and manuals, shall not exceed a two-man lift of 74 pounds (or 33.6 kilograms). If the depth of the equipment is greater than 24 inches, the maximum allowable weight shall be reduced by 33 percent.

3.5.4.3 Rackmount. Rackmount capability shall be available as an option for the equipment specified herein.

3.5.5 Manuals. The equipment shall be delivered with operator, maintenance, illustrated parts, and calibration manuals. Format and quantity shall be as specified in the contract or order. Level of maintenance philosophy, as defined in MIL-PRF-28800, shall be as specified in the contract or order.

4. REGULATORY REQUIREMENTS.

4.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement .

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

5.3 The contracting officer has the option of accepting or rejecting the product.

6. PACKAGING.

6.1 Preservation, Packing, and Marking. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of Documents.

7.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained at https://assist.daps.dla.mil or available from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

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7.1.2 The Code of Federal Regulations (CFR) may be obtained at http://www.gpoaccess.gov/cfr/index.html or available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

7.1.3 FAR and DFARS may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of the FAR may be obtained from https://www.acquisition.gov/far/ . Electronic copies of the DFARS may be obtained from http://www.acq.osd.mil/dpap/dars/dfars/index.htm .

7.1.4 Institute of Electrical and Electronics Engineers (IEEE) standards products may be obtained at http://www.ieee.org or available from IEEE Operations Center. Phone: (800)678-IEEE. Mail: IEEE Operations Center, Sales Office, 445 Hoes Lane, PO Box 1331, Piscataway, NJ, 08855-1331, USA.

7.1.5 ANSI/EIA standards may be obtained at http://www.ansi.org or http://www.eia.org or available from the Electronics Industry Association, Engineering Department, 2001 Pennsylvania Ave., N.W., Washington, D.C., 20006. Phone: 1-800-854-7179 (USA and Canada).

7.2 Ordering data. The contract or order should specify the following:

a. CID document number and revision date.
b. Product conformance provisions.
c. Packaging requirements.

7.3 Key words.

Dual-channel peak
Measuring continuous wave
Measuring peak

MILITARY INTERESTS:

Custodians:
Army – MI
Navy – SH
Air Force – 84

Review activities:
Air Force – 99
DLA - CC

Preparing activity:
Air Force – 84

Agent:
Air Force – 99

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